RE07 Global Infrastructure Investment and Management

1. Introduction to Infrastructure

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Lecture Outline

Defining infrastructure
Infrastructure and economic development
The investment need and the gap narrative
Infrastructure as an asset class
Investors, vehicles and market structure
The iron law of megaproject management
Course mechanics and the group project

Defining Infrastructure

Jochimsen (1966)

"the sum of all material, institutional and personal assets, facilities and conditions available to an economy …"

Abor et al. (2025)

Narrow: "physical assets to support economic development".

Broader: "not only the physical assets that drive economic development but also the systems and services necessary for supporting broader socio-economic activities".

Weber et al. (2016)

"all physical assets, equipment and facilities of interrelated systems and their necessary service providers offering related commodities and services to the individual economic entities or the wider public with the aim of enabling, sustaining or enhancing societal living conditions".

The three definitions agree that infrastructure is made of physical assets that serve economic activity and society. They differ on whether institutions and people belong inside the definition.

Our working definition: infrastructure is the set of physical assets, together with the systems and services that operate them, that enable economic activity and sustain living conditions in a society.

Sources: Jochimsen, R. (1966) Theorie der Infrastruktur: Grundlagen der marktwirtschaftlichen Entwicklung. Tübingen: J.C.B. Mohr, quoted in Weber et al. (2016).
Abor, J.Y., Macomber, J., Arun, T. and Murinde, V. (eds) (2025) The Routledge Handbook of Infrastructure Finance. Routledge.
Weber, B., Staub-Bisang, M. and Alfen, H.W. (2016) Infrastructure as an Asset Class: Investment Strategy, Sustainability, Project Finance and PPP. 2nd edn. Wiley.

Key Properties of Infrastructure

High sunk cost: capital is committed up front and cannot be redeployed, since a bridge has no alternative use and no resale market.

Long asset life: thirty to a hundred years or more, which pushes the valuation weight into distant and poorly observed cash flows.

Natural monopoly: minimum efficient scale often exceeds market size, so competition in the market is infeasible and regulation substitutes for it.

Inelastic demand: the service is close to essential, so volumes are less sensitive to price and to the economic cycle than for most goods.

Externalities: costs and benefits fall on parties outside the transaction, which is the standard justification for public involvement.

Political salience: the public sees what infrastructure costs and how well it performs, so governments intervene on prices and standards and the terms agreed with an investor can be reopened.

A Changing Landscape

	High sunk cost	Long asset life	Natural monopoly	Inelastic demand	Externalities	Political salience
Data centres	Yes, capital-intensive	Short, rapid technological obsolescence	**No**	Contracted, but few tenants	Heavy grid and water use	Rising, over power and water
Mobile towers	Moderate, many small sites	**Short technology cycle**	Weak, sites can be shared	Contracted operator rents	Limited	Low
Care homes and student housing	Moderate, building has other uses	Building lasts, operator may not	No	**Stable need, weak covenant**	Social benefits	High for care homes
Fibre to the home	Yes, ducts and cables	Long, electronics shorter	**Eroded by overbuild**	Close to essential	Connectivity benefits	Moderate, coverage targets
Renewable generation	Yes, fixed plant	Plant outlasts the contract	No, the grid is the monopoly	**Contracted, then merchant**	Avoided emissions	High, support schemes change
Rolling stock and aircraft	**Low, resale market**	Long, shorter than networks	No	Depends on the lessee	Through the service	Indirect

Main Sectors

Most challenging | Moderate | Least challenging

	Demand risk	Regulation	Technology	Political risk	Who pays	Track record
Transport	Severe, and network-dependen	Concessions and tolls, varies	Moderate	Fares and tolls are visible	Users, or the state if availability-paid	Long record
Energy	Low for networks, commodity price for generation	Heavy for networks, lighter for generation	Transition reshapes generation	Price caps and subsidy changes	Regulated utilities, contracted offtakers	Long, networks are the archetype
Water and waste	Close to essential, stable	Strongest monopoly, heaviest regulation	Mature	Most politically exposed pricing	Many household customers	Long record
Digital	Growing, but uncertain	Lighter than utilities	Fast change against long asset lives	Rising, over power and water use	Concentrated tenants and operators	Newest entrant, least tested
Social	No user demand risk	Contractual, not tariff-regulated	Low	Dominant risk	The paying authority's covenant is the whole credit	Established in availability PPPs

A Few Useful Taxonomies

- Economic and social infrastructure
  - Economic: serves production directly, paid for by users or through a regulated tariff
  - Social: serves public services, paid for by the state under an availability contract
- Regulated, contracted or merchant
  - Regulated: demand risk shared through the price control, low volatility
  - Contracted: the offtaker bears demand risk, bond-like while the contract runs
  - Merchant: the owner bears demand risk, equity-like and tied to the economic cycle
- Greenfield and brownfield
  - Greenfield: the asset does not yet exist, so construction risk is live and outcomes vary widely
  - Brownfield: the asset is operating, returns are lower and steadier, and most institutional capital goes here

The Ownership and Control Spectrum

Arrangement	Who owns the asset	Who bears demand risk	Example
Public provision	The state	The state	US Interstate Highway System (roads): about 48,600 miles in 2022 Thames Barrier, England (flood defence): 520 m, ten gates, protects 125 km² of London
Public corporation	The state, at arm's length	The corporation, backed by the state	Panama Canal Authority (shipping): 13,404 transits and US$5.7bn revenue in FY2025 China Telecom's Inner Mongolia park (data centres): reported largest by floor area, about 994,000 m²
Availability PPP	Special purpose vehicle, transferring on expiry	The state	Royal Children's Hospital, Australia (health): A$3.7bn of state payments over 25 years PortMiami Tunnel, US (roads): about US$32m a year in availability payments for 30 years
Concession	Special purpose vehicle, transferring on expiry	The concessionaire	Highway 407 ETR, Canada (roads): 108 km, 99-year lease sold in 1999 for about C$3.1bn Manila Water, Philippines (water): 1997 concession extended to 2037, over six million people served
Regulated private	A private company, in perpetuity	Shared through the price control	Thames Water, England (water): net debt £18.5bn against regulatory capital value £22.7bn Heathrow Airport, England (aviation): 84.5 million passengers in 2025
Full private	A private company	The owner	Switch's Citadel campus, Nevada (data centres): planned 12 million sq ft and up to 2 GW ERCOT merchant generators, Texas (power): record peak demand of 85,508 MW in 2023

Infrastructure and Economic Development

The claim: infrastructure investment raises output, and every prospectus, ministerial speech and multilateral report opens with it

Why it is hard to test: output growth generates the demand and revenue that produce infrastructure, institutions and human capital move with both, and a road in a congested corridor is not a road in an empty one

Aschauer (1989): an output elasticity of public capital of 0.39 for the United States

Calderón, Moral-Benito and Servén (2015): an elasticity of 0.07 to 0.10 across 88 countries

Ansar, Flyvbjerg, Budzier and Lunn (2016): the typical Chinese transport project fails to pay its way

Aschauer (1989): The Case for a Large Effect

Question: whether public capital raises the productivity of private output, put to annual United States data for 1949 to 1985

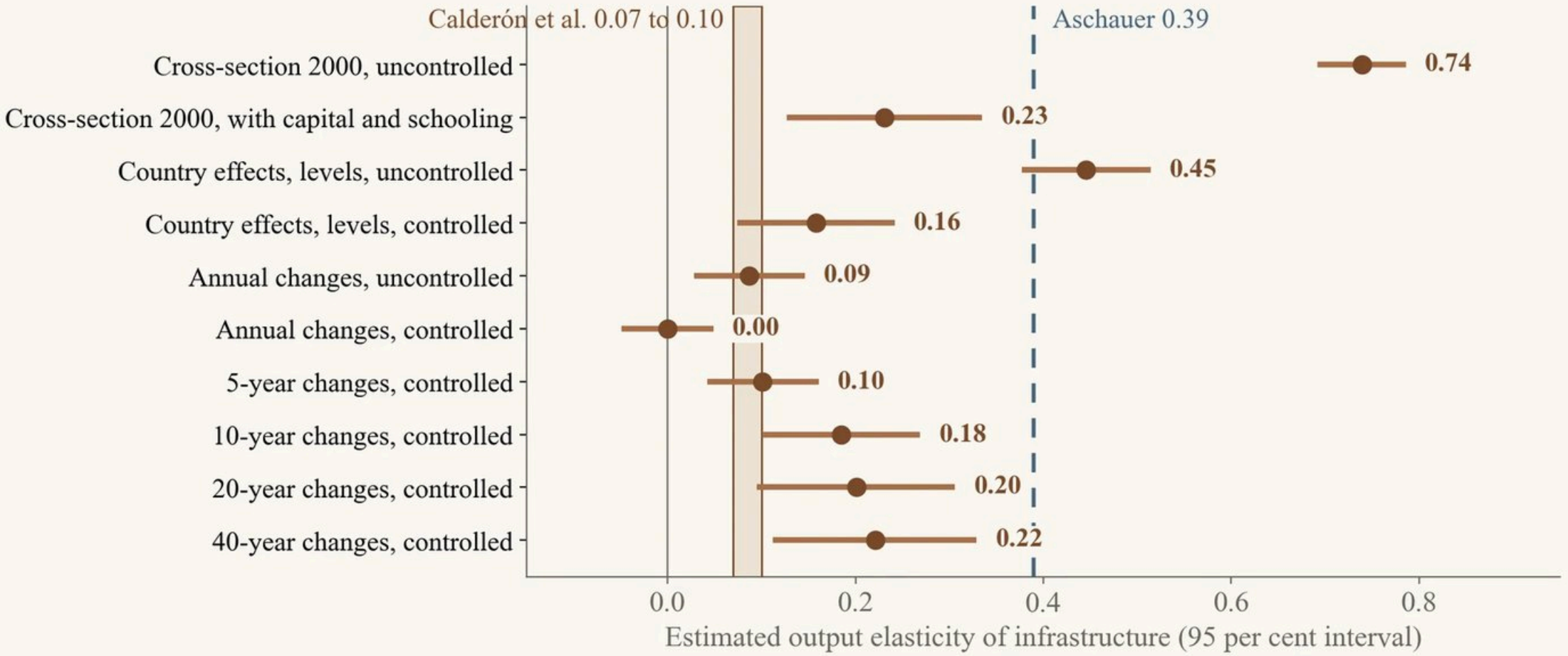
Result: an output elasticity of 0.39 for the non-military public capital stock, so a ten per cent larger stock goes with about four per cent more output

Where it sits: a core of streets, highways, airports, mass transit, sewers and water systems has the most explanatory power, and military capital has little

Reaction: an elasticity this large implies returns to public capital far above those to private capital, and the estimate was doubted almost at once

Aschauer, D. A. (1989), Is public expenditure productive?, Journal of Monetary Economics 23(2), 177–200. The elasticity is as reported in Mas, J. and Maudos, J. (2004), and the other points follow the paper's abstract.

Estimates Fall as Identification Improves



Author's estimates from the replication panel for Calderón, C., Moral-Benito, E. and Servén, L. (2015), Journal of Applied Econometrics 30(2), 177–198 (88 countries, 1960 to 2000). Benchmarks: Aschauer (1989) and Calderón et al. (2015).

Calderón, Moral-Benito and Servén (2015): Main Results

Table 3
Estimation of the Production Function
Alternative Dynamic Specifications

Column	(1)	(2)	(3)	(4)	(5)	(6)
Max # of lags	2	2	2	1	4	2
Information criterion	SBC	AIC	Imposed	SBC	SBC	SBC
Common factors	Yes	Yes	Yes	Yes	Yes	No
Physical Capital	**0.34**	**0.33**	**0.36**	**0.35**	**0.34**	**0.41**
t-ratio	35.2	30.5	22.7	31.4	32.4	33.4
hausman p-value	0.54	0.95	0.43	0.78	0.44	0.52
Secondary Education	**0.10**	**0.12**	**0.10**	**0.12**	**0.11**	**0.12**
t-ratio	15.6	14.8	8.09	18.7	17.1	16.0
hausman p-value	0.24	0.19	0.21	0.19	0.20	0.64
Infrastructure	**0.08**	**0.07**	**0.10**	**0.08**	**0.08**	**-0.02**
t-ratio	7.45	6.73	6.58	8.33	8.77	-1.49
hausman p-value	0.21	0.18	0.16	0.40	0.88	0.40
joint hausman p-value	0.44	0.38	0.24	0.25	0.45	0.85
Average R^2	0.36	0.40	0.48	0.28	0.42	0.35
Observations	3432	3432	3432	3520	3256	3432

Dependent variable is log GDP. All variables are expresed in log per worker terms. Infrastructure is an aggregate index of electricity generating capacity, main phone lines and roads. Country specific short run dynamics are either imposed or determined by information criteria (Schwarz (SBC) or Akaike (AIC)). For each regressor, the p-value from the test of the null of cross-country homogeneity is reported under the t-statistic of its respective coefficient estimate; the p-value from the joint test is reported at the bottom of the table.

Data: 88 countries, 1960 to 2000, with output, physical capital, schooling and a composite of telephones, power and roads

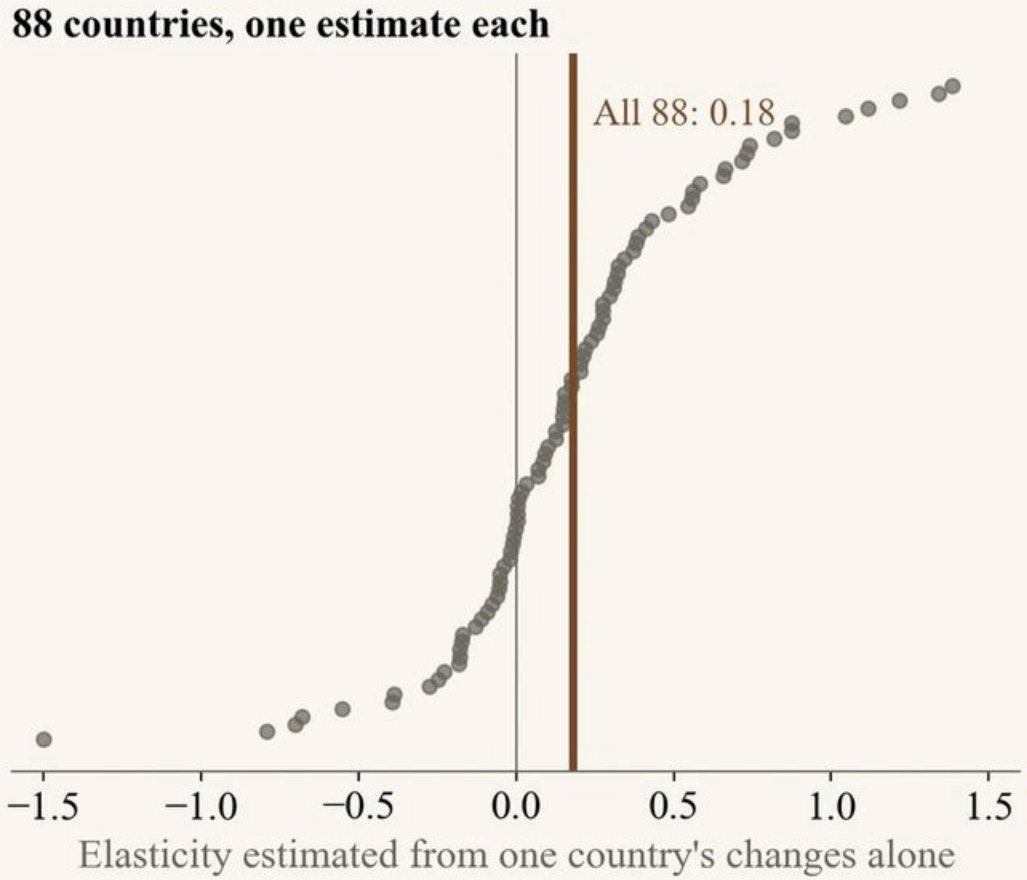
Method: a production function estimated country by country and pooled, with one common long-run slope

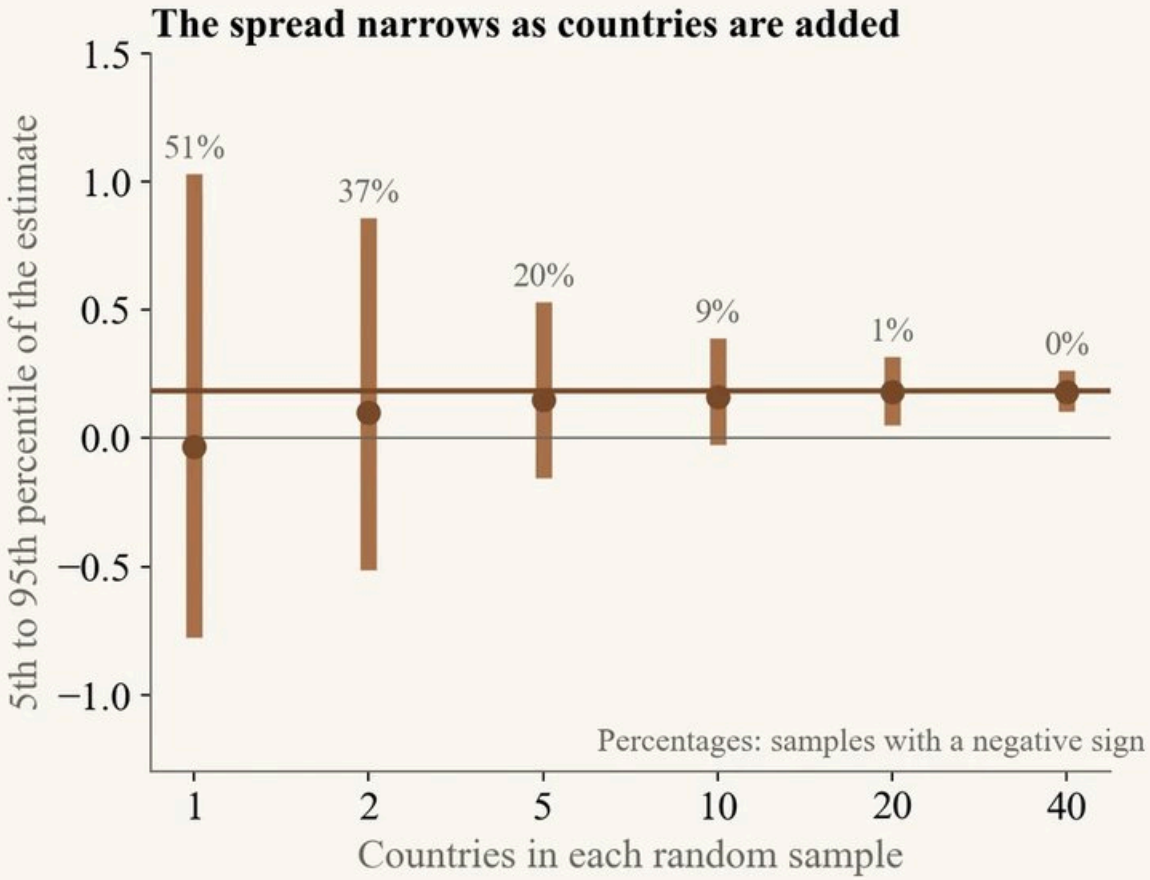
Result: an infrastructure elasticity of 0.07 to 0.10 in columns 1 to 5, with a t-ratio above 6

A warning: column 6 allows no common factors and the elasticity becomes −0.02 and insignificant

Calderón, C., Moral-Benito, E. and Servén, L. (2011), Is infrastructure capital productive? A dynamic heterogeneous approach, Banco de España Documento de Trabajo 1103, Table 3. Published as Journal of Applied Econometrics 30(2), 177–198 (2015).

From One Country to Eighty-eight

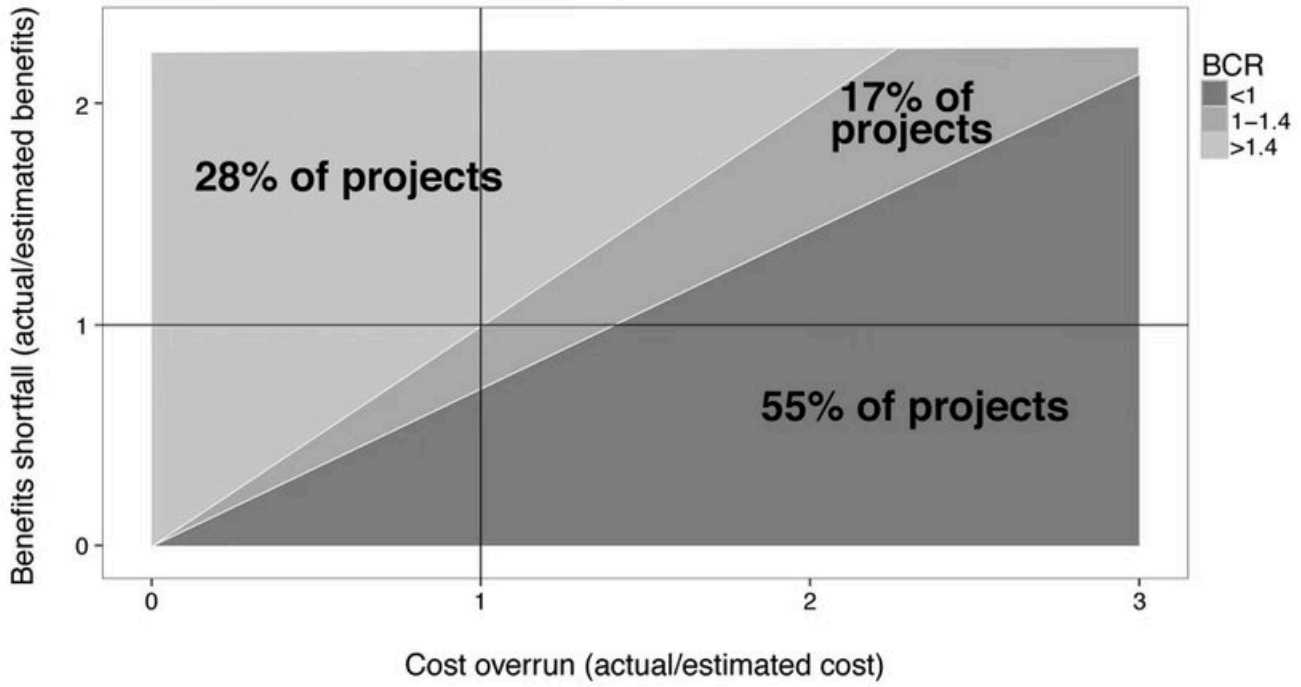




Author's estimates from the replication panel for Calderón, C., Moral-Benito, E. and Servén, L. (2015), Journal of Applied Econometrics 30(2), 177–198. Ten-year changes with physical capital and schooling held constant.

Ansar et al. (2016): Evidence from 95 Chinese Projects

Figure 6: Proportions of projects by *ex post* estimates of BCRs (n = 65)



Source: Authors' database.

Sample: 95 road and rail projects built between 1984 and 2008 across 19 provinces, worth about US$52 billion

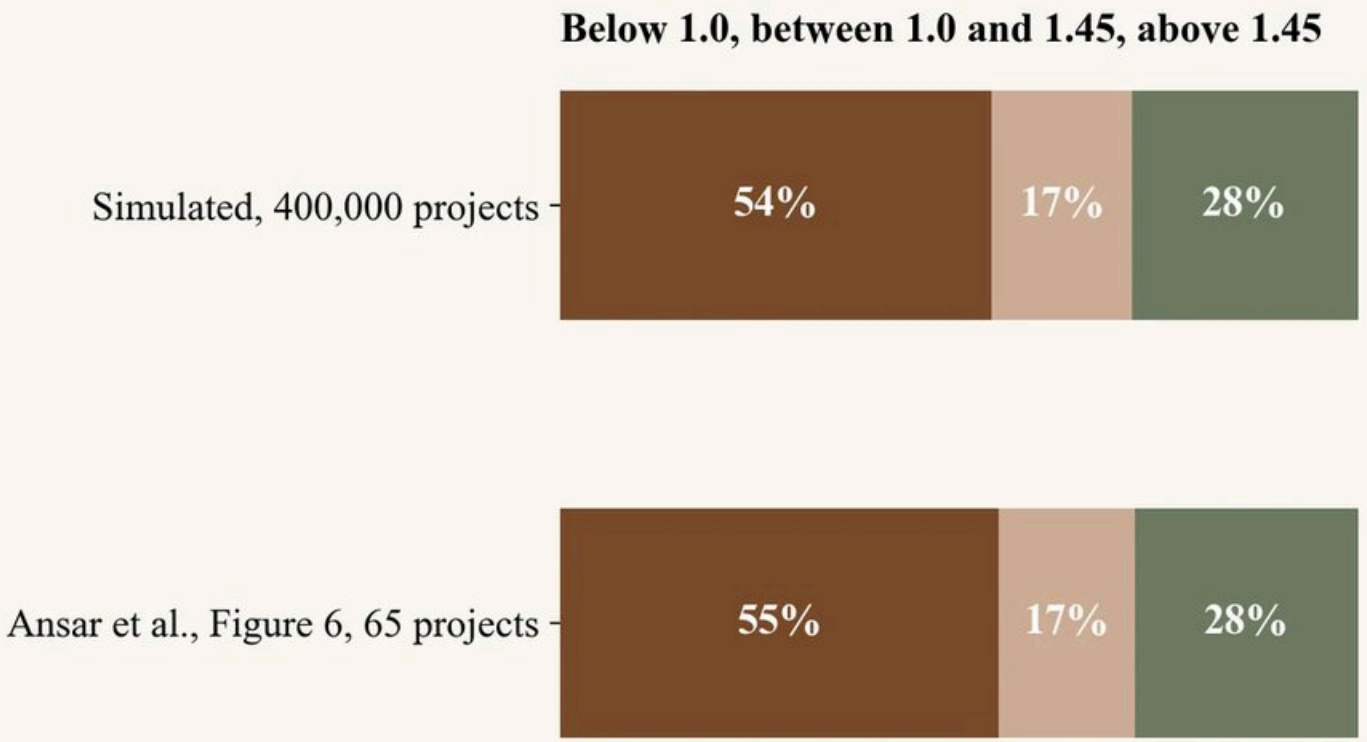
Costs: they ran over by 30.6 per cent on average and three projects in four went over budget, no better than 806 projects in rich democracies

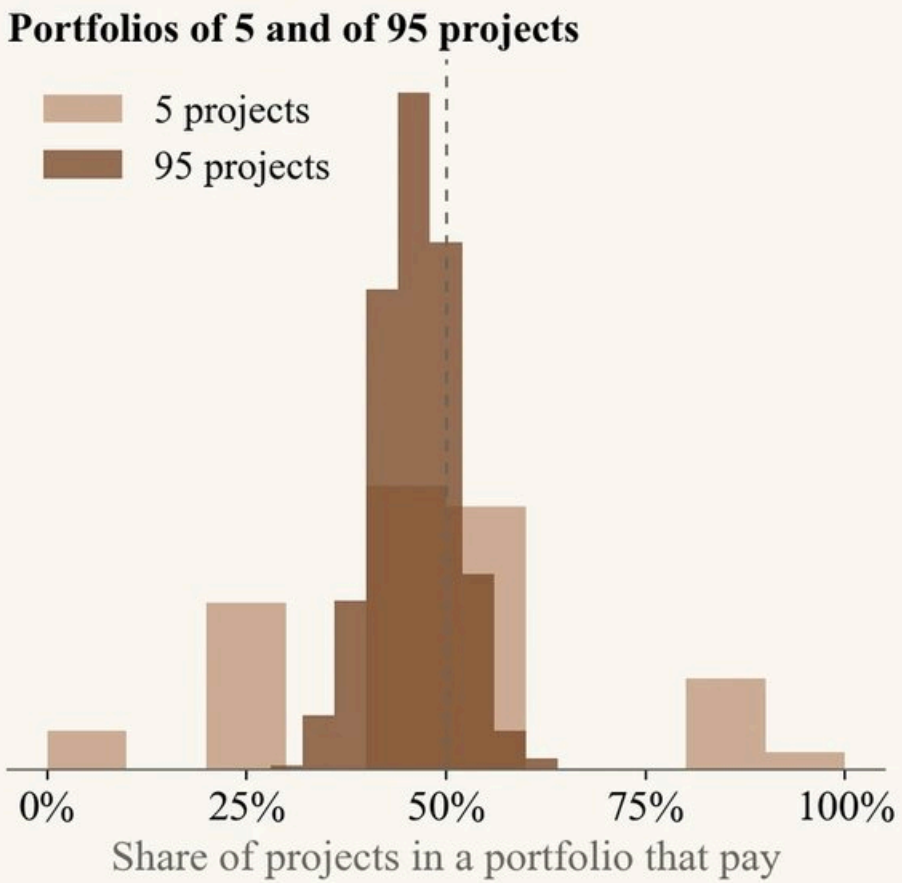
Traffic: 64.7 per cent of projects fell short, by 41.2 per cent on average, and the rest exceeded forecast by 61.4 per cent

Outcome: of 65 projects with both cost and benefit data, 55 per cent had a ratio below 1 and only 28 per cent beat the forecast of 1.4

Ansar, A., Flyvbjerg, B., Budzier, A. and Lunn, D. (2016), Does infrastructure investment lead to economic growth or economic fragility? Evidence from China, Oxford Review of Economic Policy 32(3), 360–390, Figure 6 (n = 65) and sections V and VI.

Portfolios of Projects Behave Predictably





Author's simulation using the statistics reported in Ansar, A., Flyvbjerg, B., Budzier, A. and Lunn, D. (2016), Oxford Review of Economic Policy 32(3), 360–390 (Table 3 and sections V and VI).

Summary

The claim: infrastructure raises output, and it is hard to test because governments build where growth is expected

The estimates: they fall from 0.39 to 0.07 to 0.10 as identification improves, and any single country gives a noisy answer

The projects: the typical project in the Chinese sample does not pay, even though the aggregate elasticity is positive

Newer evidence: railroads raised incomes in India (Donaldson 2018), rural roads there left consumption unchanged (Asher and Novosad 2020), a highway network cut growth in peripheral Chinese counties (Faber 2014), and United States highways earn positive but highly variable returns (Allen and Arkolakis 2022)

The implication: the case for infrastructure is a case for good infrastructure, so selection matters more than volume, which Week 2 takes up through public investment management

The Investment Need and the Gap Narrative

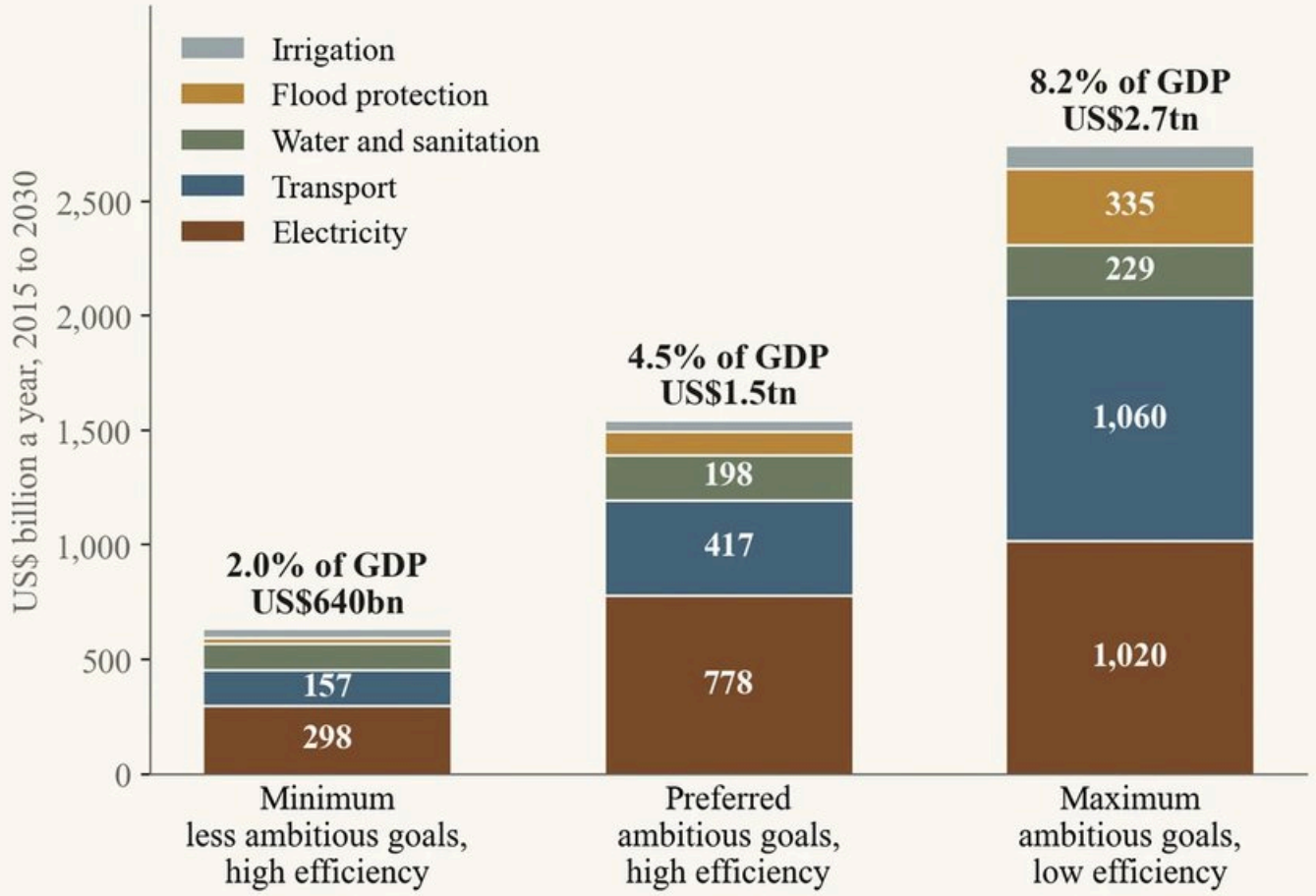
Report	Coverage	Estimated need	Gap
Global Infrastructure Hub with Oxford Economics (2017)	50 countries, seven sectors, to 2040	US$94 trillion, or US$97 trillion with the water and electricity SDGs; 3.0 per cent of global GDP rising to 3.5 per cent	US$18 trillion, almost 19 per cent of the total, unfunded on current trends
McKinsey Global Institute (2016)	The world, transport, power, water and telecoms, 2016 to 2030	US$3.3 trillion a year, 3.8 per cent of GDP	About US$350 billion a year, 11 per cent, against current spending of US$2.5 trillion
World Bank (2019), Beyond the Gap	Low- and middle-income countries, five sectors, 2015 to 2030	2.0 to 8.2 per cent of GDP a year, US$640 billion to US$2.7 trillion	No single figure: the range depends on the ambition of the goals and on efficiency
AUC and OECD (2025), Africa's Development Dynamics 2025	Africa, infrastructure, to 2040	US$155 billion a year, 5.6 per cent of GDP, to match benchmark countries' infrastructure	US$83 billion a year was invested in 2016 to 2020 (3 per cent of GDP), so the need is nearly double

Need versus gap: China has the greatest need to 2040 (US$28 trillion, 30 per cent of the world total) and the United States the largest gap (US$3.8 trillion), because a gap is need less expected spending

Comparing totals: the four estimates differ in coverage, horizon, sectors and the objective assumed, so none can be set against another without adjustment

Global Infrastructure Hub (2017), Global Infrastructure Outlook, press release, 25 July 2017; McKinsey Global Institute (2016), Bridging Global Infrastructure Gaps, as summarised by MGI; Rozenberg, J. and Fay, M. (eds) (2019), Beyond the Gap, World Bank, Overview, Figure O.1; AUC/OECD (2025), Africa's Development Dynamics 2025: Infrastructure, Growth and Transformation, pages 7, 23 and 46.

Ambition and Efficiency Drive the Estimate



Ambition: with efficiency held high, more ambitious access and quality goals lift the cost from 2.0 to 4.5 per cent of GDP a year

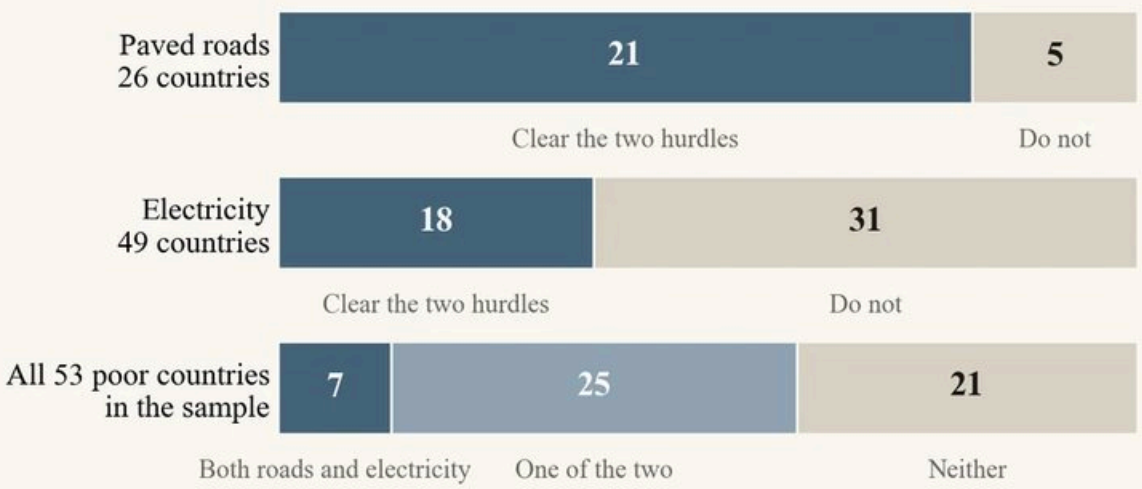
Efficiency: with ambition held, low efficiency lifts it to 8.2 per cent, as when cities sprawl and rail is built without policies to raise its use

Sectors: the swing is largest in transport (US$157 billion to US$1,060 billion) and flood protection (US$23 billion to US$335 billion)

Benchmarking: the common alternative scales up the stock that countries held at similar incomes, which carries forward past over- and undersupply

Redrawn from Rozenberg, J. and Fay, M. (eds) (2019), Beyond the Gap: How Countries Can Afford the Infrastructure They Need while Protecting the Planet (Overview), World Bank, Figure O.1 and pages 3 to 5. CC BY 3.0 IGO.

Gardner and Henry (2021): The Dual-Hurdle Test



Claim: in 2015 the World Bank argued that rich-country private capital could close the gap, earn a return and meet the Sustainable Development Goals

Test: investment is efficient and financeable from rich-country savings only if the return on poor-country infrastructure beats both poor-country private capital and rich-country private capital

Result: 7 of 53 poor countries clear both hurdles in paved roads and electricity, and 21 clear them in neither

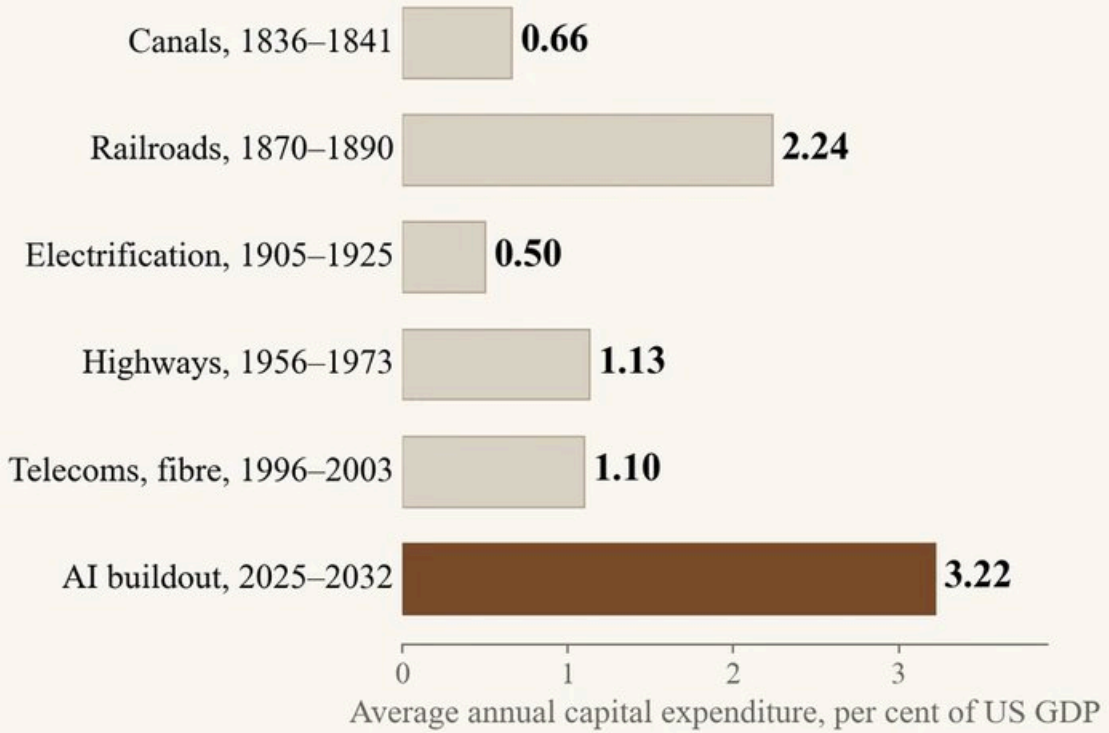
Caveat: the returns are from Canning and Bennathan (2000) and rest on 1985 data, so the result is a template for the test and not a current ranking

"Private capital is thus required to fill the gap."

The Economist, 23 July 2026, The return of the African megaproject (Middle East and Africa)

Gardner, C. and Henry, P. B. (2021), The Global Infrastructure Gap: Potential, Perils, and a Framework for Distinction, Hoover Institution Working Paper 22009, Table 1 and section 3B (pages 14 and 19 to 20). Social returns from Canning, D. and Bennathan, E. (2000).

Infrastructure Booms in History and the AI Buildout



Scale: the AI buildout would run at 3.2 per cent of US GDP a year over 2025 to 2032, above the railroads at 2.2

Pipeline: 63 GW operates, 494 GW is in the pipeline, and the central scenario completes 188 GW by 2032

Caution: the scenario assumes 43 per cent of the pipeline's capacity is never completed

Lesson: booms turn costly when demand falls short, financing tightens or debt outlives the technology

Van Nieuwerburgh, S. (2026), Financing the AI Buildout, NBER Working Paper 35865, October 2026, Table 1 (page 5) and pages 4 and 13. Earlier episodes as reported in the paper; the AI figure is a scenario.

Summary

The estimates: published gaps range from US$18 trillion unfunded to 2040 (Global Infrastructure Hub) to 2.0 to 8.2 per cent of GDP a year in low- and middle-income countries (World Bank), because each answers a different question

The method: a gap is the cost of a chosen objective at an assumed efficiency, so the ambition and efficiency assumed matter as much as the data

The inference: a gap is not a pipeline of investable projects, and on the one returns dataset available only 7 of 53 poor countries cleared both hurdles

Week 2: who chooses the objective and how projects are selected, through institutional frameworks and stakeholder governance

Weeks 3 and 4: why private capital does not simply fill the gap, and how risk allocation and financing structure make a project bankable

Weeks 6 and 8: whether infrastructure earns the returns that need implies (Week 6), and what climate and sustainability goals add to the cost (Week 8)

Infrastructure as an Asset Class

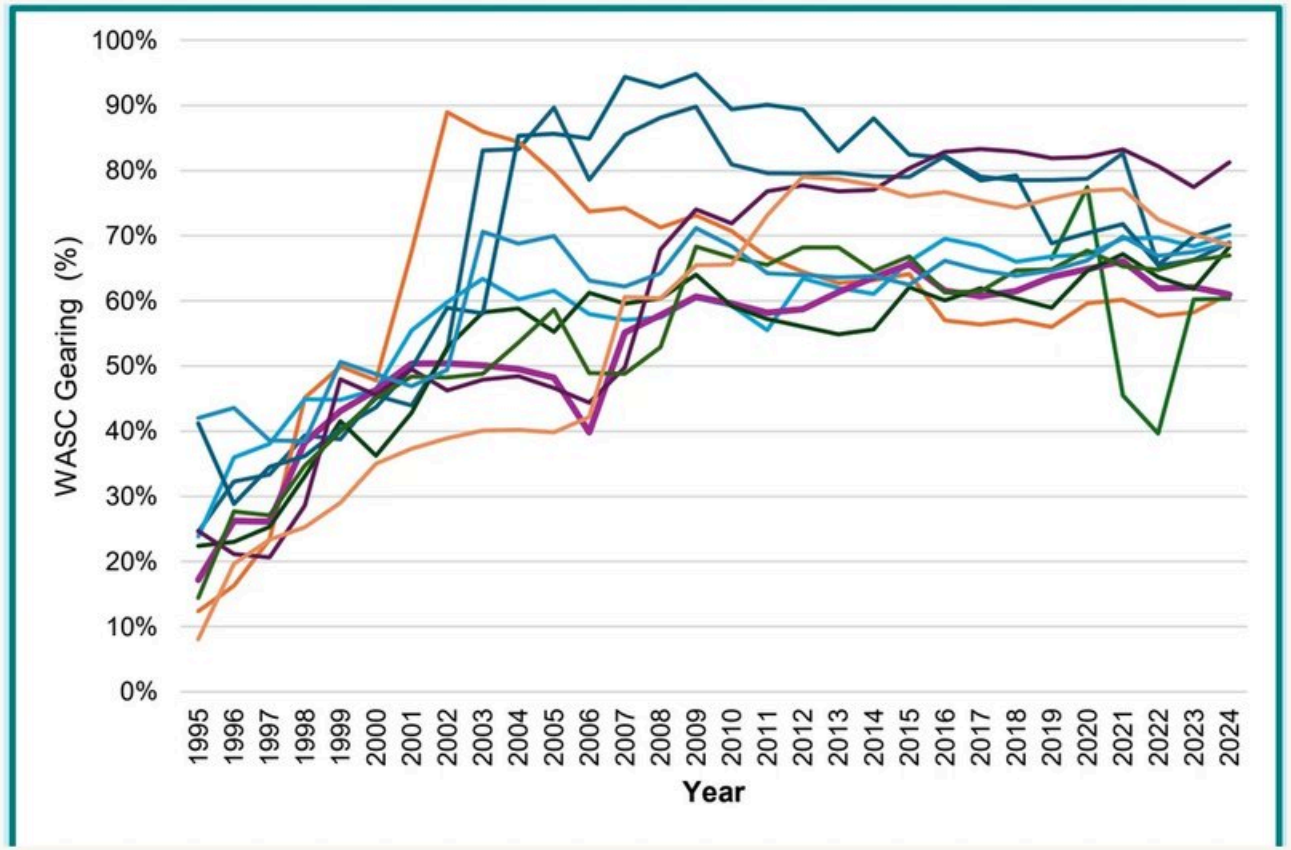
The test: an asset class is a group of investments whose returns share common drivers, behave alike under stress and are not reproduced by equities, bonds and real estate

The five claims: stable cash flows, inflation linkage, low correlation with equities, long duration, and a premium for illiquidity

The common structure: every claim is a claim about a contract or a regulatory settlement, so each is tested against that contract and not against the concrete

The evidence: one regulated utility, Thames Water, tests the first two claims through the Independent Water Commission and company filings, and one academic study tests the funds through which investors hold such assets

Thames Water: Stable Revenue, Fragile Equity



Gearing: water companies raised debt against regulatory capital from an industry average of 4 per cent in 1991 to 72 per cent in 2009, and Thames Water, the dark purple line, has been above 80 per cent for most of the last decade

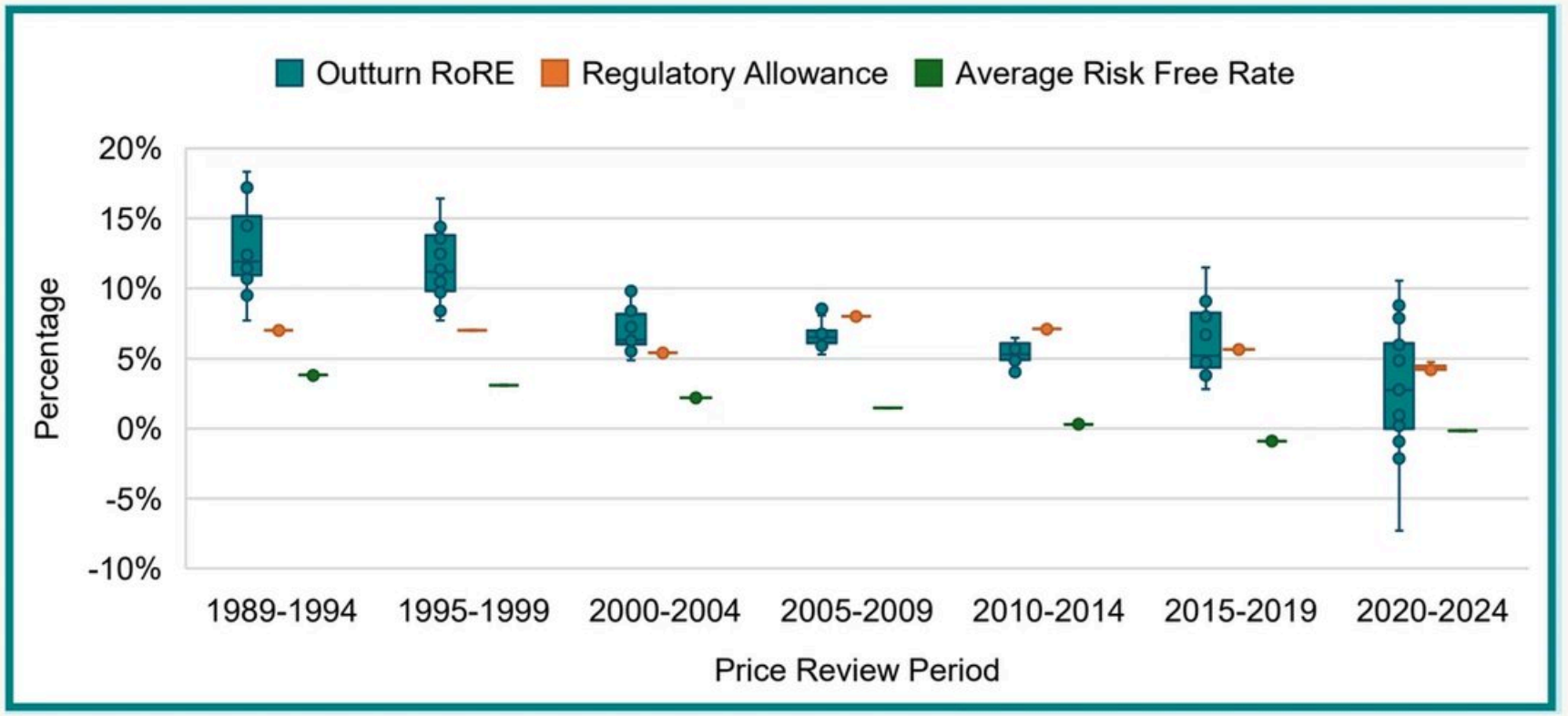
Thames Water: at March 2024 its regulated group held net debt of £16.1 billion, 80.6 per cent of regulatory capital value, and 87.4 per cent including the holding company's debt

The outcome: its parent defaulted on £1.4 billion of debt in April 2024, and in June 2025 a private equity bidder withdrew after ten weeks of due diligence

The lesson: the revenue is stable by design, but the equity is what remains after debt service, so stability does not reach the shareholder

Independent Water Commission (2025), Final Report, 21 July 2025, Figure 24 and Box 41 (pages 330 and 331), using Ofwat data; Thames Water Utilities Limited (2024), Investor Report 31 March 2024, page 29; The Economist, 9 April 2024 and 5 June 2025.

The Allowed Return and the Realised Return



Source:

Independent Water Commission (2025), Final Report, 21 July 2025, Figure 22 and Box 40 (pages 322 and 323), the Commission's analysis; the comparator returns are as reported to the Commission by the Global Infrastructure Investor Association.

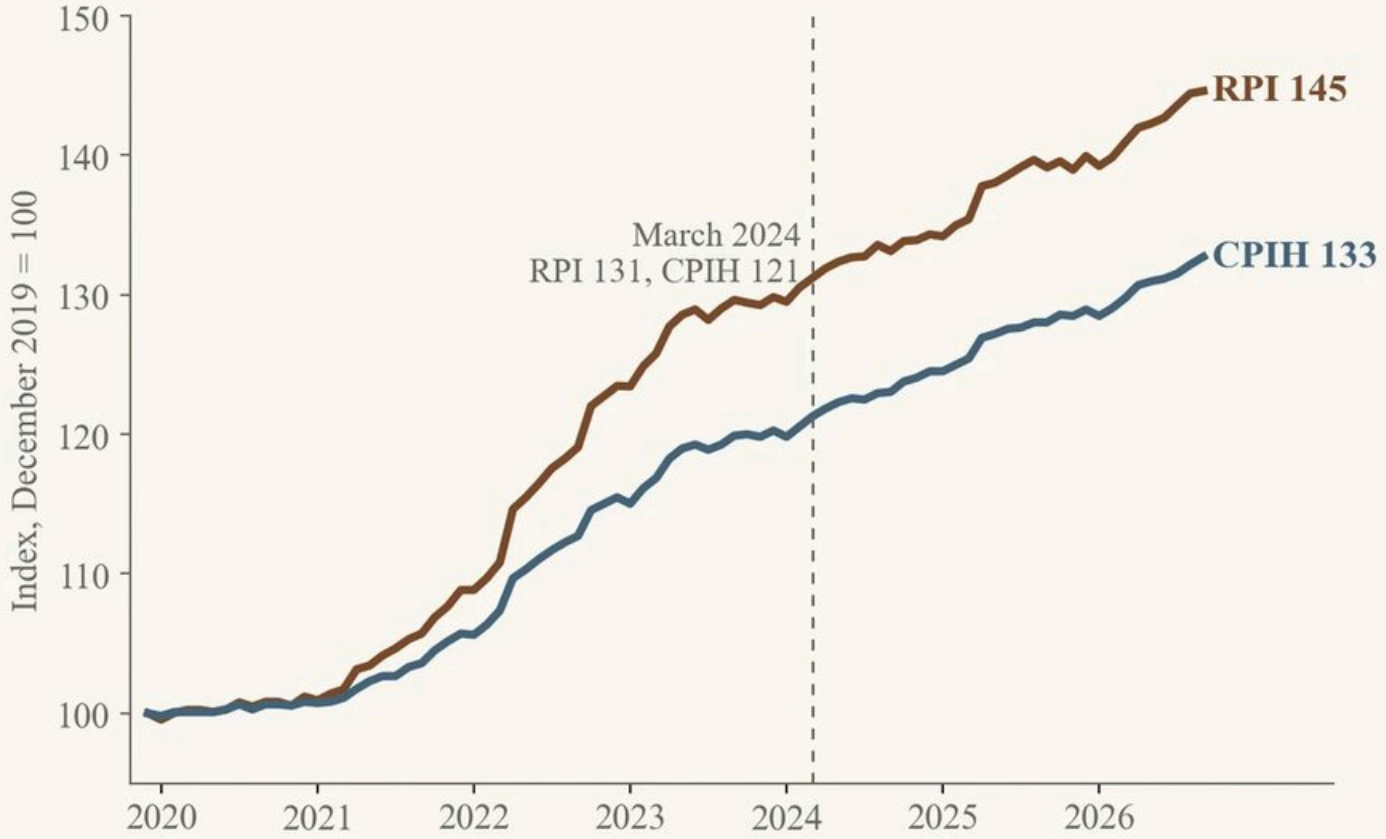
Average returns: fell from around 13 per cent in 1989 to 1994 to around 3 per cent in 2020 to 2024 on comparable measures

Dispersion: the gap between the best and the worst performers widened, and in 2020 to 2024 the median outturn sat below the regulatory allowance

The risk-free rate: some returns fell below it, and Price Review 2019 was the first time any water company earned less than the risk-free rate

Comparators: investors told the Commission that comparable regulated markets offer 8 to 12 per cent on equity, against 7.2 per

Inflation Linkage Is Contractual



Revenue: Thames Water's bills rise with CPIH, as The Economist reports

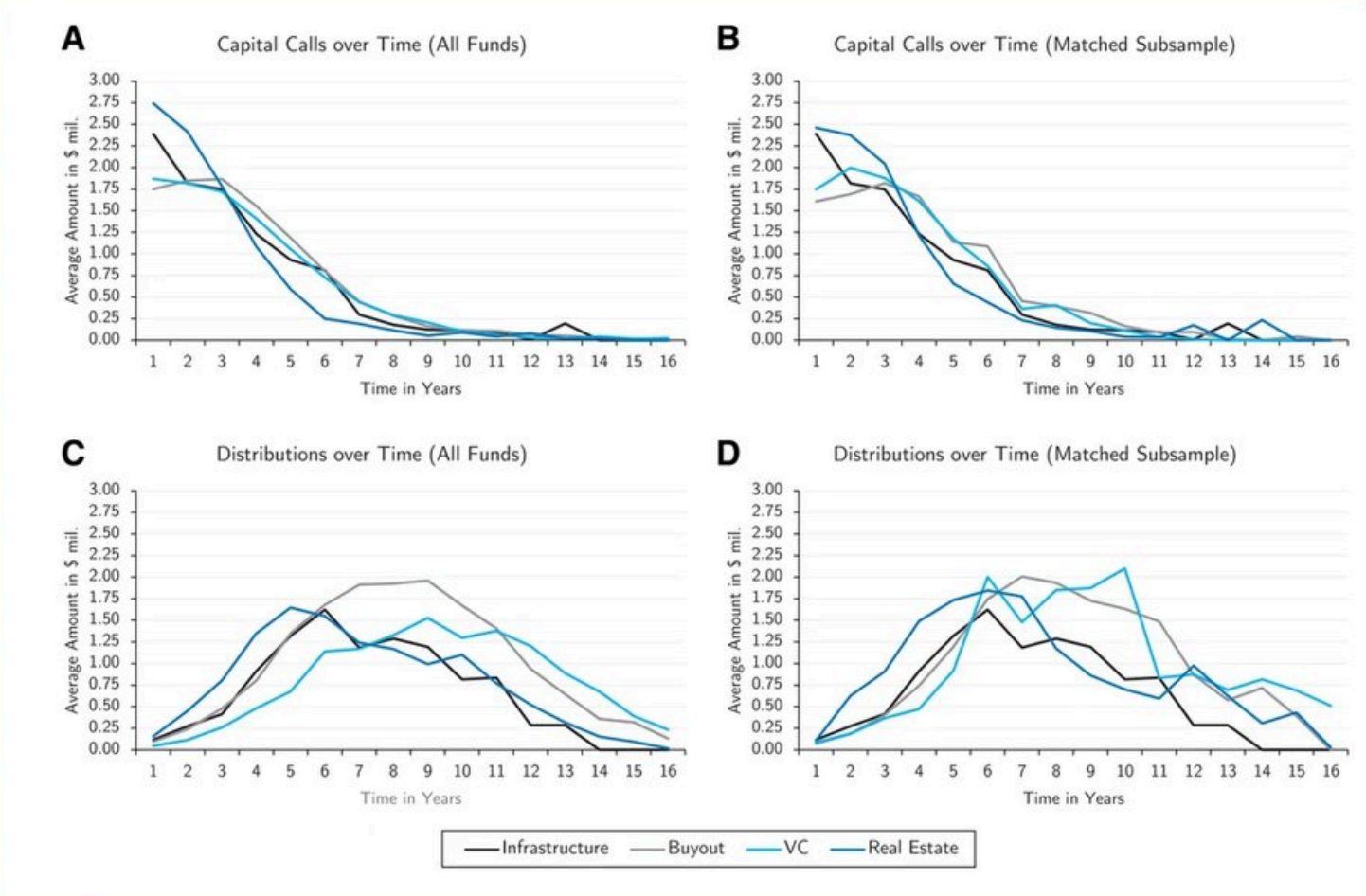
Debt: 55.7 per cent of its senior debt is index-linked, to RPI, so its financing costs follow a different index from its revenue

The gap: from December 2019 to March 2024 RPI rose 31.2 per cent and CPIH 21.3 per cent, and by September 2026 the gap was 11.9 percentage points

The test: which index, applied to what share of revenue, with what lag, and until when

Office for National Statistics, RPI all items index (CHAW) and CPIH all items index (L522), monthly, retrieved 25 October 2026; Thames Water Utilities Limited (2024), Investor Report 31 March 2024, page 29; The Economist, 9 April 2024.

Funds Are Not the Asset



The vehicle: institutions reach infrastructure mainly through closed private funds of finite life, although the assets they hold last for decades

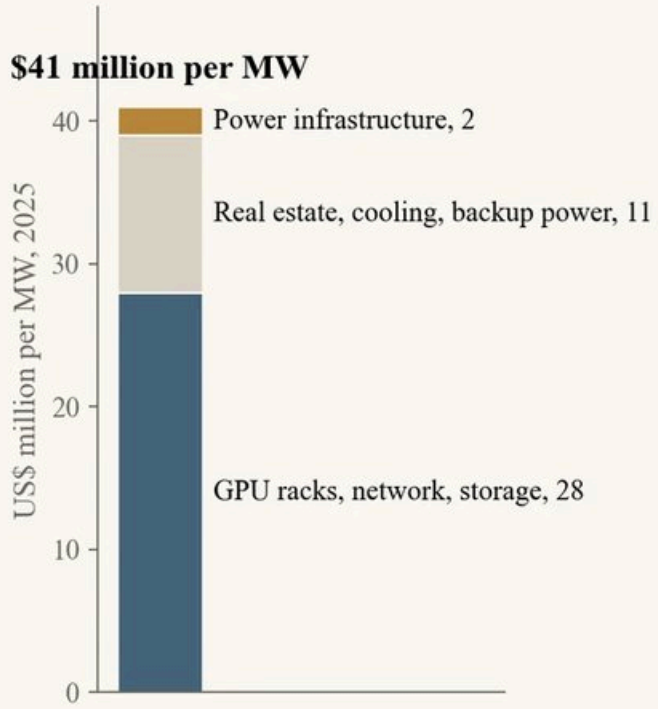
The cash flows: Andonov, Kräussl and Rauh reject the hypothesis that these funds deliver more stable and diversifying cash flows than other private funds, and find them procyclical and driven by quick exits

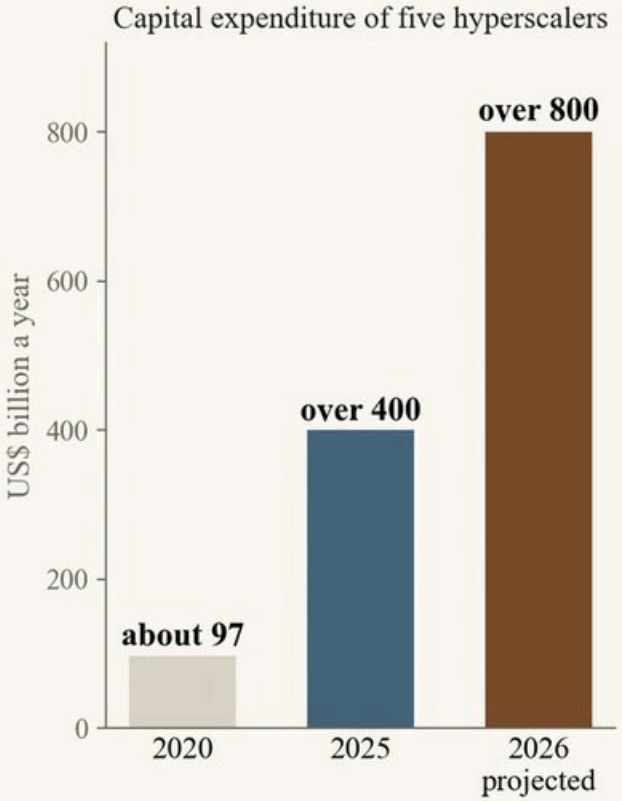
Tail risk: returns on listed United States infrastructure, reconstructed back to 1927, carry tail risk similar to a broad portfolio of United States equities

The premium: an illiquidity premium is the extra return required to hold an asset that is costly to trade, and it is hardest to earn when liquidity disappears

Andonov, A., Kräussl, R. and Rauh, J. (2021), Institutional Investors and Infrastructure Investing, The Review of Financial Studies 34(8), 3880–3934, Figure 3 (page 3900) and page 3882; Bianchi, R. J., Bornholt, G., Drew, M. E. and Howard, M. F. (2014), Journal of Banking and Finance 42, 314–325; The Economist, 7 November 2019.

The Cost of a Gigawatt of AI Capacity





Cost: $41 million a megawatt, so $41 billion a gigawatt, with two thirds of it computing equipment

Lives: buildings and power systems last for decades, chips for far fewer years, and the debt may outlive the chips

Cash: capital expenditure of five hyperscalers rose from $97 billion in 2020 to over $400 billion in 2025 and passes their operating cash flow in 2026

Wrapper: leases, joint ventures, project debt and private credit move leverage off the tenant's balance sheet

Van Nieuwerburgh, S. (2026), Financing the AI Buildout, NBER Working Paper 35865, October 2026, pages 2 to 7 and 13, and Figure 1 on page 6. Capital expenditure bars for 2025 and 2026 show the lower bound of 'more than'.

Summary

The strong claim: fails, because returns are not uniformly stable, the stability of revenue does not reach the equity, and the funds that hold the assets behave like private equity

The weaker claim: survives for assets held directly, with a regulated or contracted revenue base and a long holding period, which can give long-duration, partly indexed cash flow, and identifying which assets qualify is most of the job

Week 2: the economic regulation and price control that set the allowed return, and the regulator's own risk

Week 3: who holds the assets and why: more than four fifths of infrastructure fund assets are in Western markets while the need is greatest elsewhere, which echoes the dual-hurdle test of the previous section

Weeks 5 and 6: the fund manager and the intermediary (Week 5), then performance and valuation, including appraisal smoothing and public market equivalents (Week 6)

Investors, Vehicles and Market Structure

Liabilities: an investor allocates because of what it owes: an inflation-indexed pension liability is matched by regulated infrastructure, while an endowment with no such liability needs a return premium to buy it

Pension regimes: in Britain, defined benefit schemes in surplus have no reason to take illiquid risk, and defined contribution schemes face a 0.75 per cent fee cap and daily dealing that limit unlisted assets

Public investors: in the data of Andonov, Kräussl and Rauh they earn a net IRR 1.8 percentage points lower than private investors in infrastructure funds

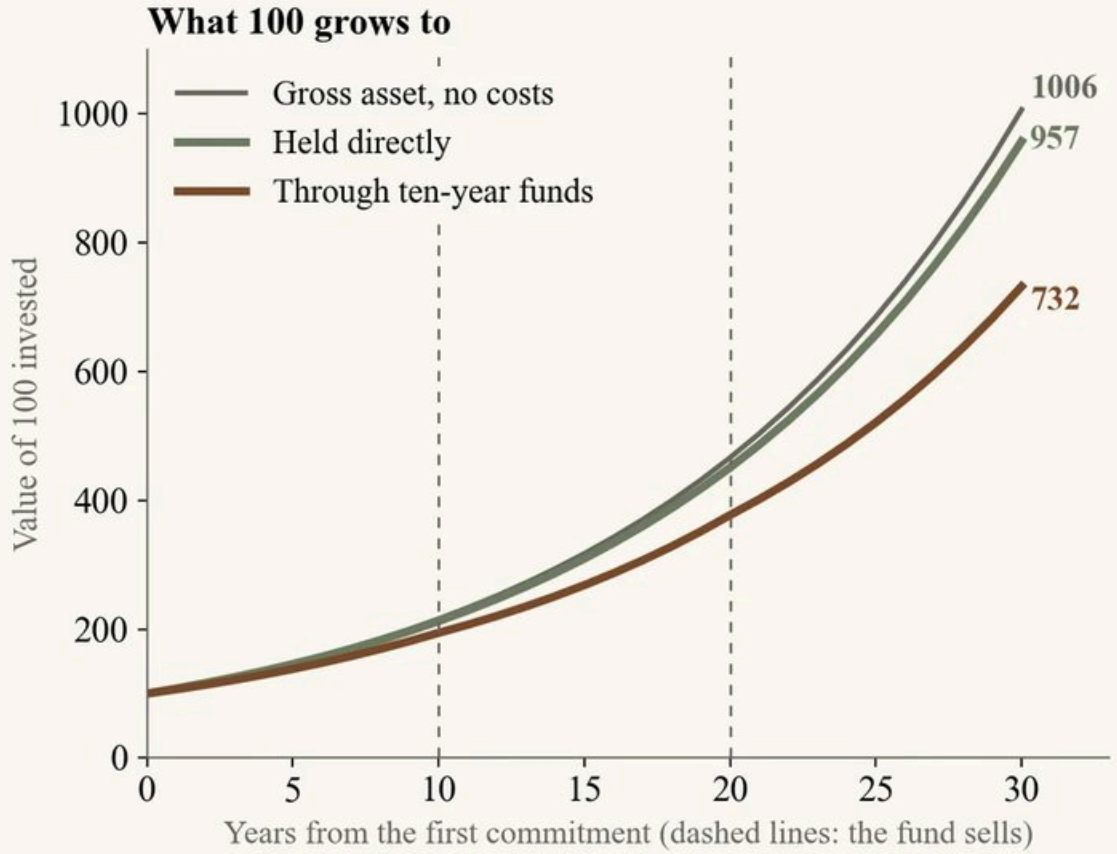
Concentration: fund assets rose almost five-fold in a decade to US$1.3 trillion, with over four fifths in Western markets, while the need is most pronounced in the global south

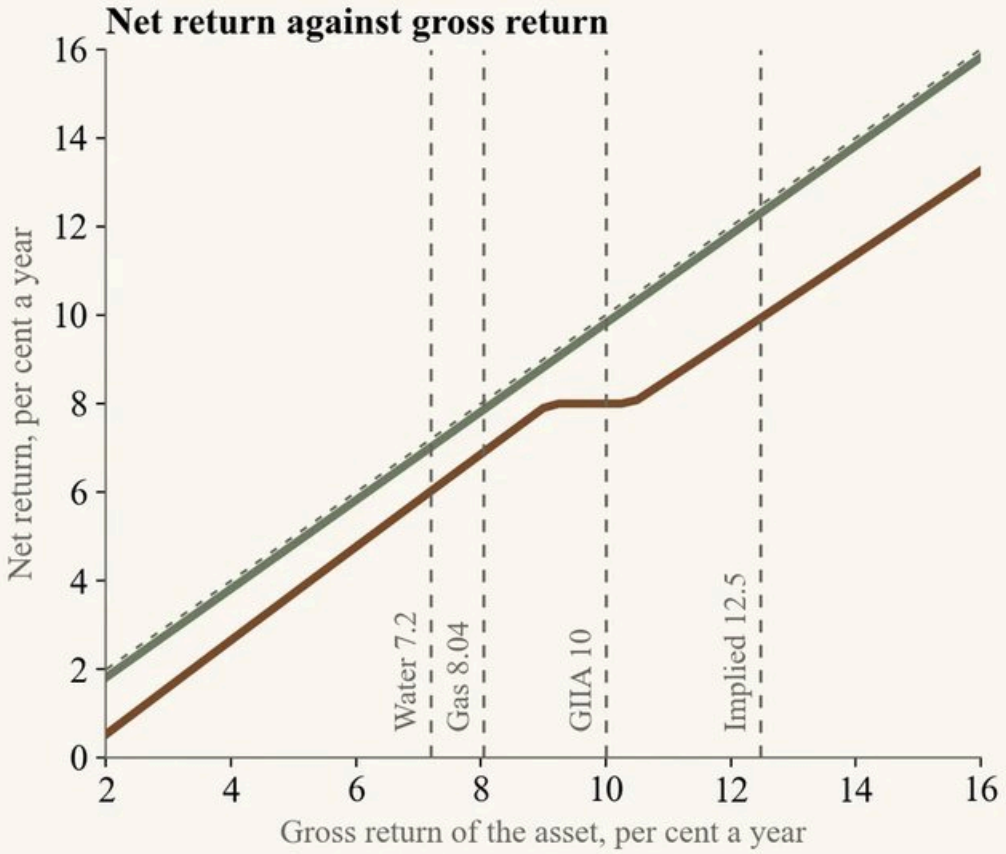
Routes into the Asset and the Cost

Route	Control	What it costs	Typical holder
Listed equity	None	Low, and no performance fee	Any investor
Closed-end fund	None, subject to fund governance	1.5 to 2.0% a year on commitments, and 15 to 20% of profits above a hurdle	Mid-sized pension funds and insurers
Fund of funds	None	3.06% of net assets a year in one SEC-filed example, before the carry of the funds below	Individuals through brokers, institutions in the cheapest class
Co-investment alongside a fund	Limited	Reduced or no fees	Large pension funds
Direct investment	Full, or shared with partners	An internal team: 0.18% a year on average for large Canadian funds, all assets	Sovereign wealth funds, large plans, corporates
Infrastructure debt	None, but covenants and step-in rights	Low	Insurers, debt funds, banks

ILPA (2021), What's market in fund terms, pages 4 and 12; StepStone Private Infrastructure Fund, SEC Form 424B3, 6 March 2025, page 11; Beath, A. D., Betermier, S., Flynn, C. and Spehner, Q. (2021), The Journal of Portfolio Management, Investment Models 2021, page 2.

A Worked Example of Fee Drag





Author's model, run on the terms in the text (fee_model.py); asset returns from Independent Water Commission (2025), Final Report, page 323; observed net IRR from Andonov, Kräussl and Rauh (2021), Review of Financial Studies 34(8), Table 3, page 3897; fee terms from ILPA (2021) and a StepStone prospectus (SEC Form 424B3, 2025).

The Direct Model and Its Limits

In-house: large Canadian funds manage 80 per cent of their assets in-house, against 34 per cent for non-Canadian funds above US$50 billion, and in-house management cuts costs by about one third

Cost: they spend 18 basis points on internally managed portfolios against 7 for peers, but 121 on external managers against 86, across all asset classes and with carried interest included

Scale: the model works best at scale, yet small Canadian funds that cannot replicate it have raised their in-house share and their infrastructure allocation

The match: direct ownership removes the fund life, which is typically 10 to 12 years, so the holding period can match the asset, and consortium ownership then raises its own governance problem

Summary

Who: an investor's liabilities decide what it buys, and capital concentrates in a few large investors and in brownfield assets in stable markets

The price: a closed-end fund charges 1.5 to 2.0 per cent a year and 15 to 20 per cent of profits above a hurdle, and a fund of funds layers a second set of fees on the first

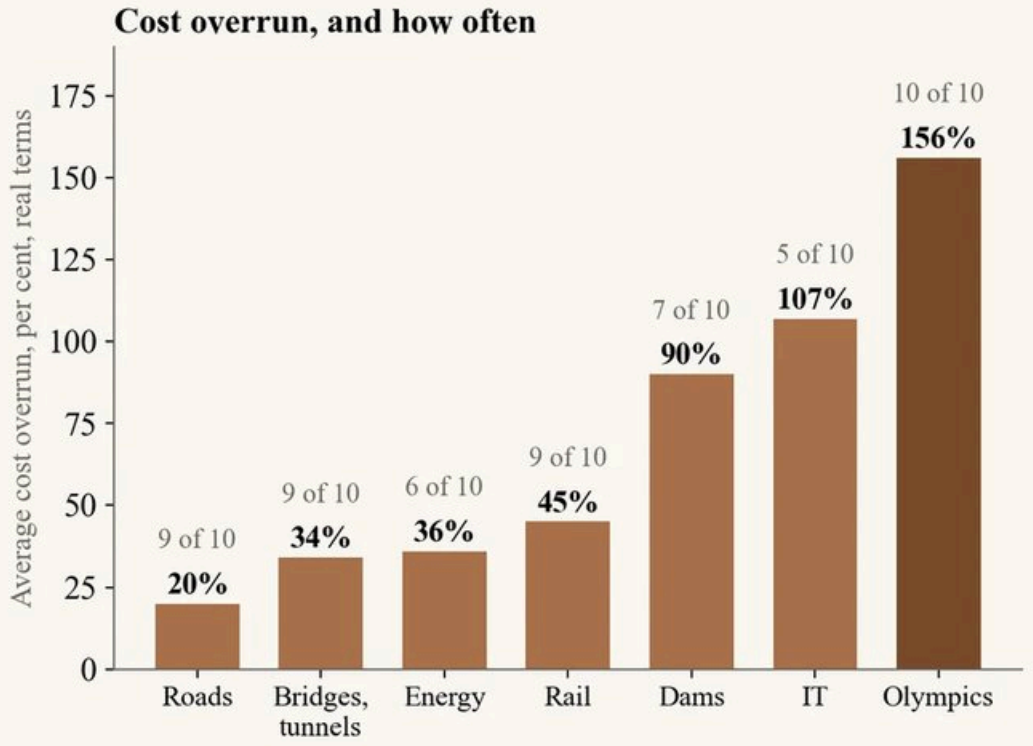
Fee drag: on a regulated asset earning 8 per cent, ten-year funds cost about 114 basis points a year, and a fund must earn 12.5 per cent gross to deliver the 9.9 per cent net observed

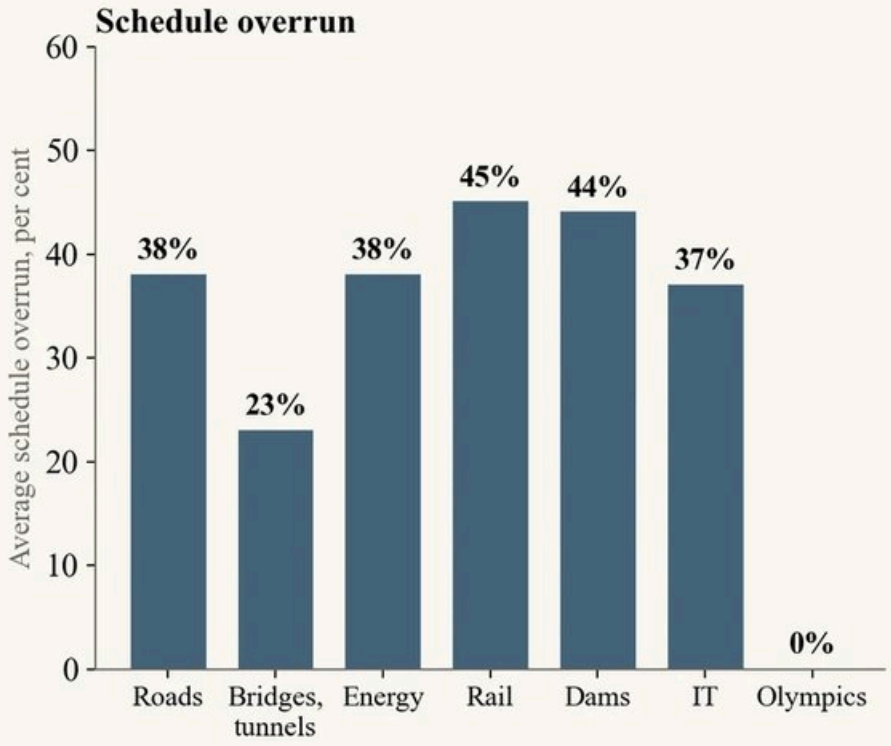
Week 3: why capital does not reach the need, and how development finance institutions mobilise private capital

Weeks 4 and 5: consortium governance and shareholders' agreements (Week 4), and the fund manager, consolidation and the strategic corporate owner (Week 5)

Week 6: performance measurement: the public market equivalent and the J-curve

The Iron Law of Megaprojects





Source:

Flyvbjerg, B., Budzier, A. and Stewart, A. (2016), The Oxford Olympics Study 2016, Saïd Business School Research Papers RP 2016-20, Table 4 (page 15)

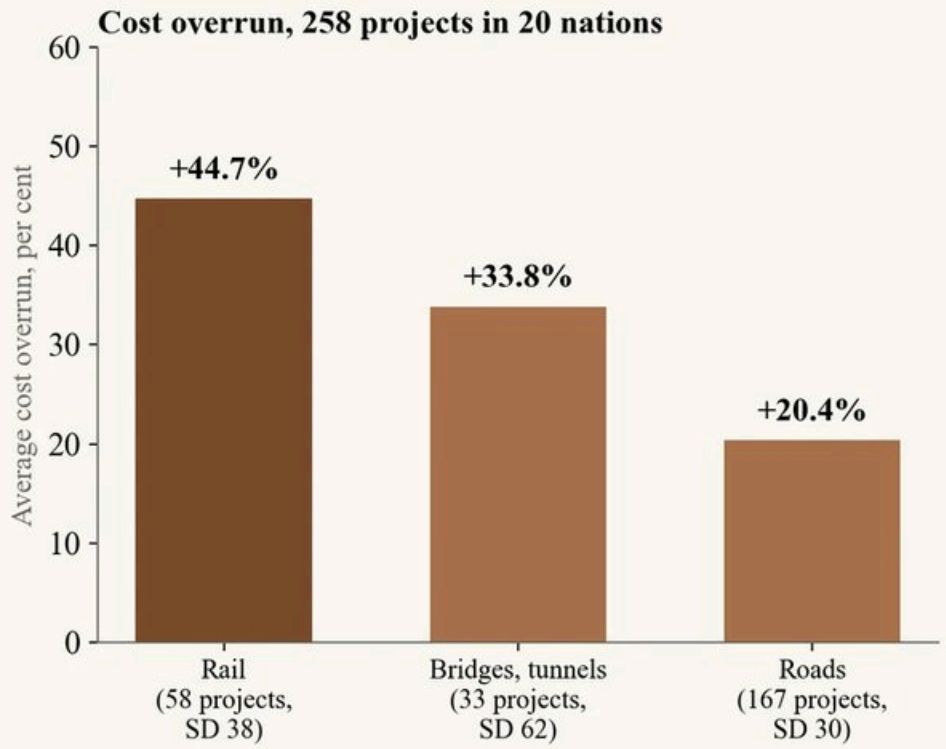
Flyvbjerg, B. (2014), Project Management Journal 45(2), 6–19.

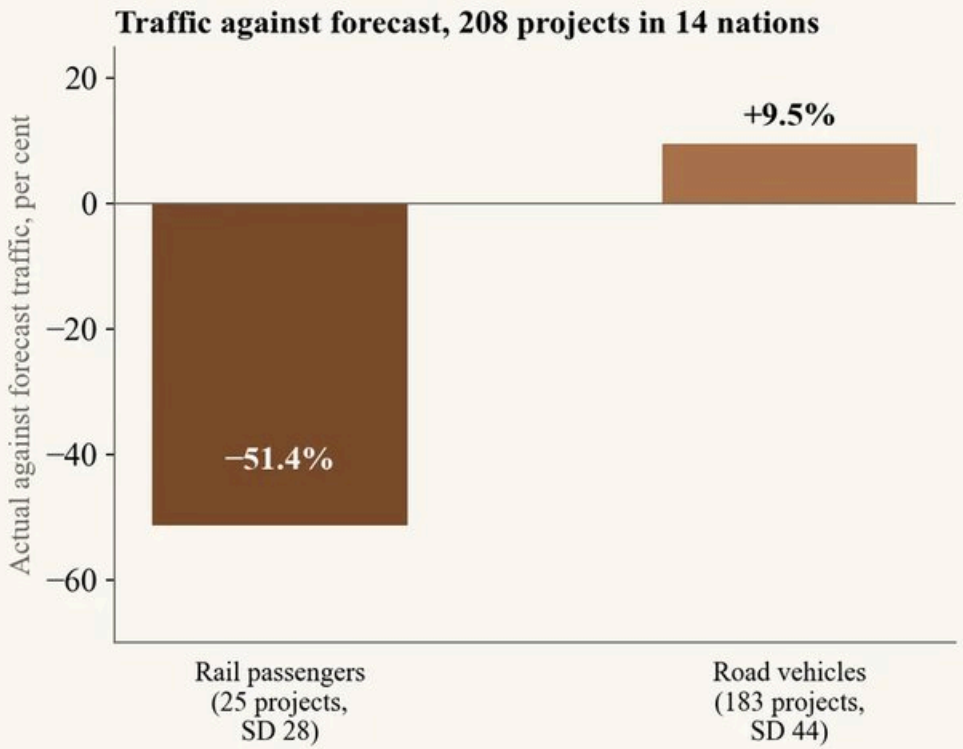
The law: over budget, over time, over and over again: nine in ten megaprojects overrun their budgets, and overruns have not fallen over the seventy years for which comparable data exist

The scale: global megaproject spending is estimated at US$6 to 9 trillion a year, about 8 per cent of world output

Cost: average overruns run from 20 per cent for roads and 45 for rail to 90 for dams, 107 for IT and 156 for the Olympic Games

Forecasts Err in a Predictable Direction





Source:

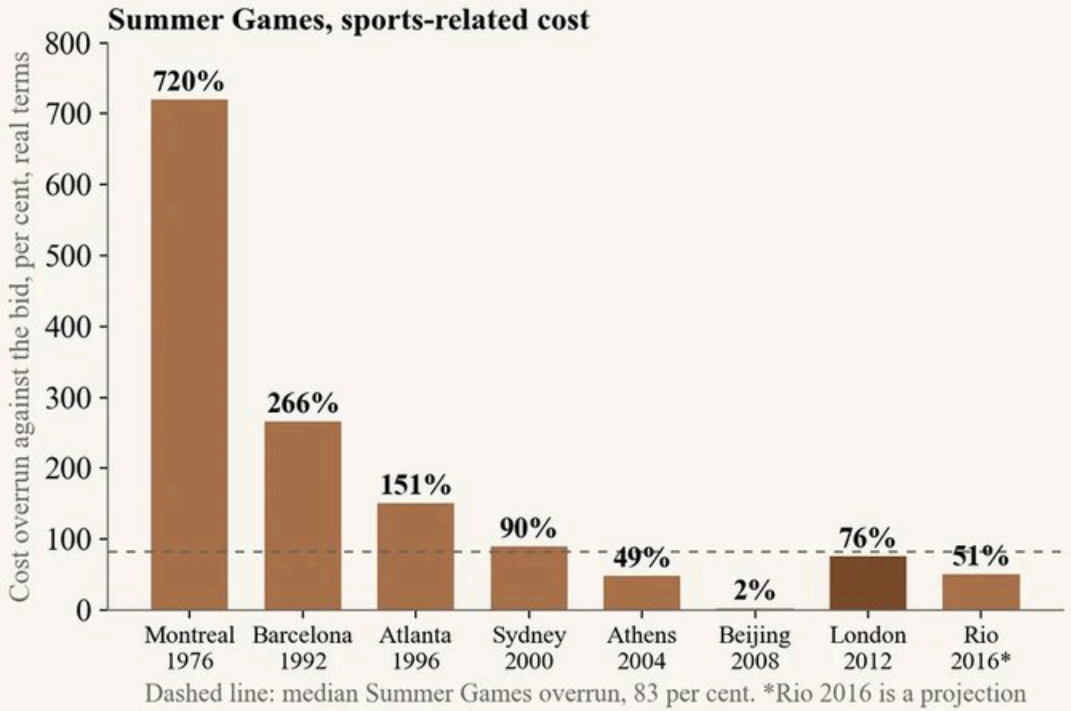
Flyvbjerg, B. (2009), Oxford Review of Economic Policy 25(3), 344–367, Tables 1 and 2, from Flyvbjerg, Holm and Buhl (2002, 2005), Journal of the American Planning Association 68(3) and 71(2).

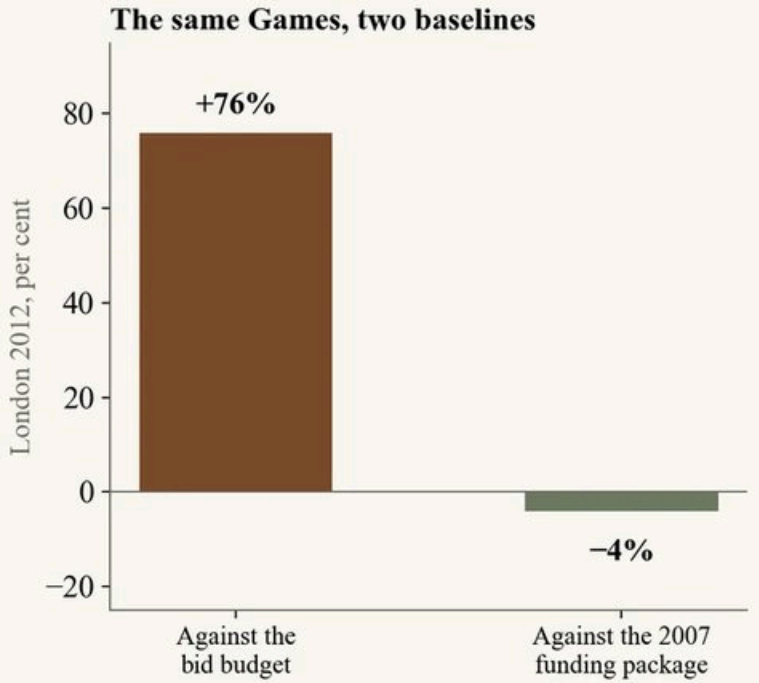
Cost: in 258 transport projects in 20 countries, rail overran by 44.7 per cent, bridges and tunnels by 33.8 and roads by 20.4, in constant prices from the decision to build

Demand: rail traffic came in 51 per cent below forecast, which is a forecast 106 per cent too high, while road traffic erred by 9.5 per cent on average with a wide spread

Why: technical error would be centred on zero and shrink over time, and neither holds, so the data point to optimism bias and to strategic misrepresentation

London 2012: Delivery under a Fixed Deadline





Source:

Flyvbjerg, B., Budzier, A. and Stewart, A. (2016), The Oxford Olympics Study 2016, Tables 1 to 3 (pages 9 to 12);

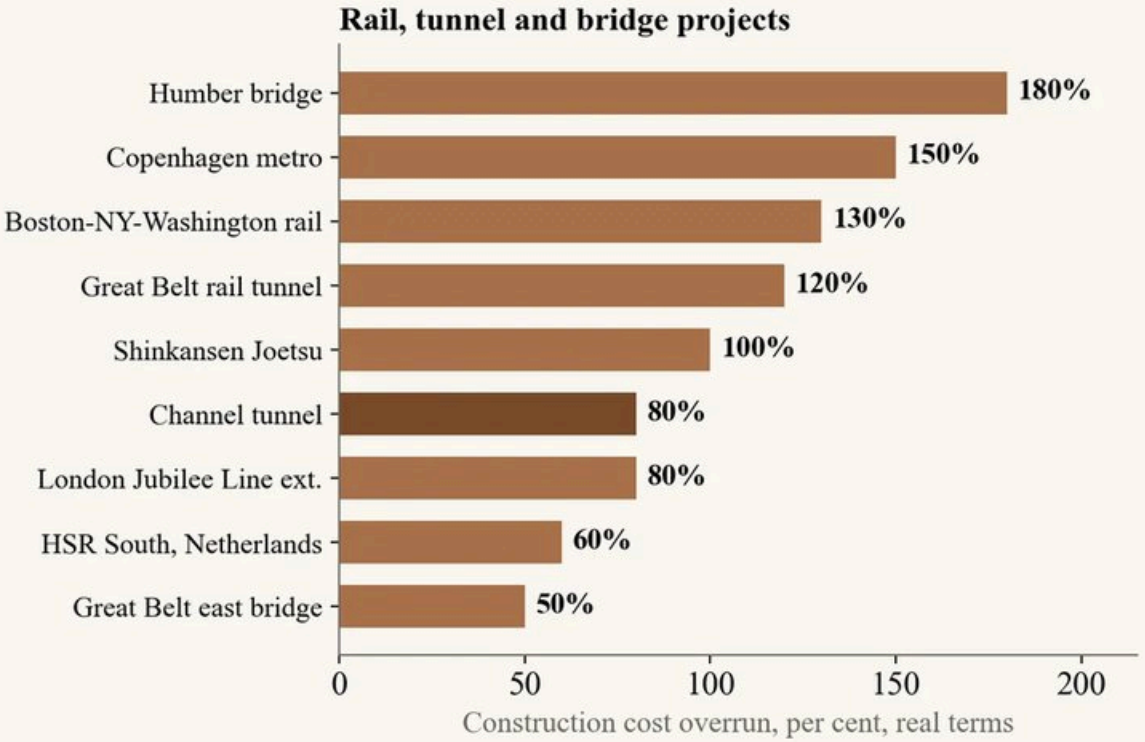
Department for Culture, Media and Sport (2012), London 2012 Olympic and Paralympic Games Quarterly Report, October 2012, pages 9 and 14.

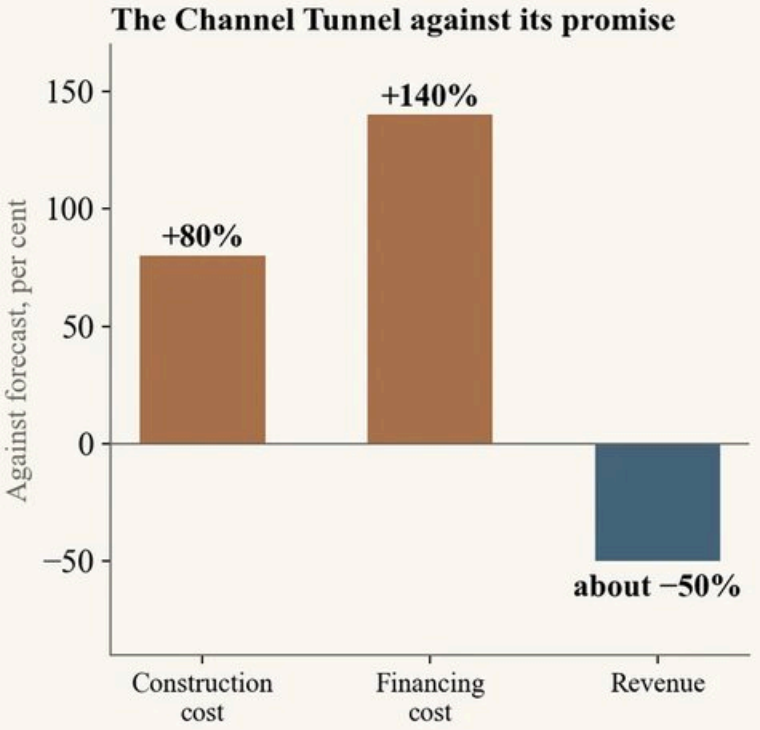
The deadline: no Games in the Oxford data has overrun its schedule, yet every Games with data overran its budget, and London finished all main venues by July 2011, a year ahead

Two baselines: London cost 76 per cent more than the bid budget in real terms, but £377 million, or 4 per cent, less than the £9,298 million funding package set in 2007

Other Games: Montreal 1976 overran by 720 per cent, Barcelona 1992 by 266 and Atlanta 1996 by 151, against a median of 83 for Summer Games

The Channel Tunnel: Asset against Investment





Source:

Flyvbjerg, B. (2014), What you should know about megaprojects and why, Project Management Journal 45(2), 6–19, Table 2 and pages 9 and 10;

Anguera, R. (2006), Transportation Research Part A 40(4), 291–315.

The promise: investors were told that 10 per cent would be a reasonable allowance for unforeseen circumstances in construction costs

The outcome: construction cost 80 per cent more than planned and financing cost 140 per cent more, in real terms from the decision, and revenue came in at half the forecast

The verdict: an internal rate of return of minus 14.5 per cent and a loss to the British economy of US$17.8 billion, with investors bearing part of it when Eurotunnel was restructured

Lecture Plan

Week	Topic	Applied case, second hour	Guest
1	Introduction to infrastructure	Team formation and project launch, with a conversation with a guest	Central Bank of the UAE
2	Institutional frameworks and stakeholder governance	Heathrow expansion and the consenting regime	Department for Transport
3	Private investment in infrastructure	A local impact fund and a public pension scheme	Newcore Capital
4	Infrastructure financing and deal structuring	The UK experience of PFI and privatisation, topic to be agreed	Former Minister for Transport
5	Intermediation and deal brokerage	Pricing an operating regulated asset in a sale	Principal investor
6	Performance and valuation	How infrastructure indices are built and how unrated project loans perform	Central Bank of the UAE
7	Technology and asset life	Building and financing a data-centre platform	Fund manager
8	Sustainability, ESG and emerging trends	Water infrastructure and the environmental dimension of ESG	Central Bank of the UAE
Extra	Presentations to the Global Infrastructure Advisory Panel	Group project presentations	Panel members

Assessment

Group project report, 50 per cent: teams analyse a live infrastructure investment issue supplied by a panel member and submit strategic and financial recommendations as a written report

Presentation to the panel: the recommendation is presented to the Global Infrastructure Advisory Panel in an extra session at the end of the module (date to be confirmed), and the project is assessed on analysis, judgement under uncertainty and the quality of the recommendation

Written examination, 50 per cent: individual work, two hours, written in person

What the two test: the project tests judgement on a live question over several weeks, and the examination tests each student's own command of the module's tools under examination conditions

What a Good Project Report and Presentation Look Like

A recommendation: state what should be done, in the first paragraph of the report and the first minute of the presentation, and defend it against the strongest objection

A number: every recommendation has a financial consequence, so quantify it, state the assumptions and show what would change the answer

A risk allocation: name who bears each material risk under your proposal and why that party is the efficient bearer of it

An honest range: give the base case and the distribution around it, because a single point estimate presented without a range will be marked down

A stakeholder map: identify who must agree, what each party needs, and where the proposal is politically vulnerable

A defence: the panel will test the recommendation, so prepare the answers to its hardest questions as carefully as the case

Reading for Week 2

Delmon: chapter 5 on the institutional framework and chapter 6 on selecting transactions

Hart (2003): Incomplete contracts and public ownership, The Economic Journal 113(486), C69 to C76; seven pages, and the theoretical spine of the week

Engel, Fischer and Galetovic (2013): The basic public finance of public-private partnerships, Journal of the European Economic Association 11(1), 83 to 111

IMF: Public Investment Management Assessment Handbook, chapters 1 and 2, for the fifteen institutions of public investment management

Optional: Flyvbjerg (ed.), chapter 18 on the Norwegian quality assurance regime, which is the working example of what the iron law section argues is needed

Where the Module Connects to the Rest of the MPhil

Week	Real Estate Finance	Planning, Growth and Regeneration	Environmental Policy
1	RE01 (MT): real estate as an asset class	PGR03 (MT): location theory and agglomeration	EP02 (MT): market failure; EP06 (LT): energy resources
2	RE01 (MT): WACC as a regulator's allowed return	PGR01 (MT): UK planning; PGR04 (MT): institutions and rent-seeking	EP02 (LT): asymmetric information; EP07 (LT): adjudication
3	RE02 (LT): private vehicles, fees and risk	PGR05 (LT): financialisation; PGR07 (MT): regeneration finance	EP06 (LT): the valley of death; EP01 (LT): climate finance
4	RE01 (MT): capital structure; RE03 (LT): funding	PGR01 (LT): land value capture	EP02 (MT): instrument choice; EP02 (LT): liability rules
5	RE02 (MT): information and market efficiency	PGR07 (MT): the three industry seminars	EP07 (LT): private governance
6	RE02 (LT): valuation methods and testing	PGR02 (MT): policy evaluation; PGR05 (LT): transport appraisal	EP02 (LT): valuation and evaluation; EP06 (LT): multicriteria analysis
7	RE02 (LT): net zero and climate risk	PGR05 (LT): the green transition; PGR01 (LT): planning for net zero	EP06 (LT): throughout; EP01 (LT): climate litigation

Summary

Defining infrastructure
Infrastructure and economic development
The investment need and the gap narrative
Infrastructure as an asset class
Investors, vehicles and market structure
The iron law of megaproject management
Course mechanics and the group project